Deciding D-stability is coNP-hard

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Abstract

A real square matrix A is D-stable if DA is Hurwitz for every positive diagonal matrix D. Explicit criteria are known up to order three, and decision procedures for order four. The general decision problem has long been regarded as computationally difficult. Deciding D-stability of a rational matrix is coNP-hard, and deciding whether some positive diagonal scaling has an eigenvalue in the open right half-plane is NP-hard.

The reduction is from PARTITION. For weights $w_1, \dots, w_m$ it produces an integer matrix of order $m + 4$ with entries of polynomial bit length. The matrix is D-stable if the weights admit no equal split, and has a strictly destabilizing positive scaling otherwise. The hardness therefore holds for the promise problem and for D-semistability as well. It is hardness in the weak sense (binary encoding), since the entries grow like the fourth power of the total weight. The hard matrices consist of a 4×4 core together with m first-order attachments at one coordinate.

The mechanism is a subset-sum obstruction. Fast attachments act as static load and slow ones are invisible, so the matrix is destabilized exactly when a subset sum of the loads lands in a window of loads at which the core is D-unstable. The window must be exponentially narrow and yet exactly certified. It is obtained from a core whose window is born at the irrational parameter $\sqrt{3}$, so that Pell approximations p/q of $\sqrt{3}$ give rational windows of width $\Theta(1/q)$. Every imaginary-axis contact is controlled by one polynomial inequality that places its height under the window's semicircle.

The mathematical argument, including the explicit integer encoding, is formally verified in Lean 4 with Mathlib. Only the complexity-theoretic packaging (running time and trivial instances) is informal.

Keywords: D-stability; diagonal scaling; computational complexity; coNP-hardness; subset sums; Pell equation; Routh–Hurwitz; formal verification.

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1 Introduction

A real $N \times N$ matrix A is *D-stable* if DA is Hurwitz, that is, has all eigenvalues in the open left half-plane, for every positive diagonal matrix D. Equivalently, the linear system $\dot{x} = DAx$ is asymptotically stable whatever the positive time scales of its coordinates. Since DA and $AD = D^{-1}(DA)D$ are similar, one may scale on either side; we use right scaling from Section 2 on.

The notion goes back to Enthoven and Arrow [15] and Arrow and McManus [1] in the stability of competitive equilibria. It has since become a standard object of matrix theory [26, 12, 22, 32, 24, 29]. Explicit criteria are classical for $N \leq 3$ [7], and decision procedures are known for $N = 4$ [27, 25]. For general N no explicit criterion is known, and recent work develops hierarchies of sufficient tests [29, 30].

Chen, Fan and Yu [10] related D-stability to a real structured singular value. They remarked that checking D-stability "may in general be an NP-hard problem". The remark has been repeated since [20]. Recent work describes exact tests as computationally intractable for larger systems [30], and computational methods for D-stability as a largely open area [9]. As far as we are aware, no reduction has been given (Section 8). The known hardness results for structured singular values [6], for interval matrices [34, 33] and for matrix polytopes [21] concern general matrices, interval families or convex hulls. Hardness for general instances does not transfer to the special matrices through which D-stability is expressed.

Main result. Let PARTITION be the NP-complete problem [28, 19]: given positive integers $w_1, \ldots, w_m$, decide whether the index set splits into two parts of equal weight. Matrices are encoded by their rational entries in binary.

Theorem. *There is a polynomial-time algorithm that maps every PARTITION instance $w = (w_1, \ldots, w_m)$ to an integer matrix $M(w)$ such that:*

- *if w is a no-instance, then $M(w)$ is D-stable;*
- *if w is a yes-instance, then some positive diagonal scaling of $M(w)$ has an eigenvalue with positive real part.*

Consequently:

- *deciding D-stability of a rational matrix is coNP-hard, and so is deciding D-semistability;*
- *deciding whether a rational matrix admits a destabilizing positive diagonal scaling is NP-hard.*

All three statements hold already for the promise problem that separates D-stable matrices from strictly D-unstable ones. For nontrivial instances, $M(w)$ has order $m + 4$ and consists of a 4×4 core with m first-order attachments at one coordinate.

The entries of $M(w)$ have magnitude $\Theta(T^4)$, where $2T = \sum_j w_j$. This is not bounded by a polynomial in the input length, so the hardness is in the weak sense, for binary encodings (Section 8).

The mechanism. The reduction uses *star matrices*. A core B interacts with m scalar attachments through one coordinate, the *port*. Attachment j relaxes toward the port on its own time scale and feeds back with load r_j. An attachment much faster than the port acts as a static load on the port's diagonal, and one much slower is invisible; this is developed in a companion manuscript [13], and the facts needed here are proved in Section 2. Hence, if the core loaded statically by a subset sum $\sum_{j \in S} r_j$ is D-unstable, so is the star matrix. If the set of D-unstable static loads is a window $\mathcal{U}$, instability of the star matrix looks like the question "does some subset sum lie in $\mathcal{U}$?".

There are two obstacles to turning this into a reduction:

(i) The window must be exponentially narrow compared with its distance from the ends of the load range, since otherwise only boundedly many large pieces matter and the subset question is easy. At the same time it must have exactly known, rational position and width.

(ii) Avoiding the window must be *sufficient* for D-stability. The attachments of a fixed partition could still produce contacts, that is, imaginary-axis eigenvalues, at intermediate time scales.

How the obstacles are overcome. We use a 4×4 core whose principal minors are equal within each order and type (inner coordinates versus minors containing the port). Its scaled characteristic polynomial then depends on the three inner rates only through their elementary symmetric functions. Its Routh–Hurwitz test can fail only through a single quadratic $K(h)$ in the port's self-coefficient h (Section 3).

In our one-parameter family ρ, the window $\{K < 0\}$ is born at the irrational point $\rho = \sqrt{3}$. Pell solutions $p^2 - 3q^2 = 1$ therefore give windows with rational endpoints and width $\Theta(1/q)$, which settles (i).

For (ii) we prove one polynomial inequality. It shows that the normalized height of every contact of the static core lies under the semicircle spanned by the window. The total lag of the attachments, on the other hand, is at least the distance from their effective static load to the nearest subset sum. Together these exclude all contacts. A continuation in ρ from a window-free core then gives D-stability.

The proof is self-contained. The safety theorem (Theorem 2.5) is proved here, and the stable direction uses only elimination, the elementary rounding bound of Lemma 4.5 and continuation. The companion paper [13] supplies motivation and a more general theory of star matrices, which the hardness proof does not require.

Verification. The proofs of Sections 2–5 are formally verified in Lean 4 [14] with Mathlib [38]. The exceptions are explicitly marked: Remarks 3.4 and 3.6, which are not used in the proofs and are checked by exact computation, and Corollary 5.2. The formal root is `DStabilityHardness.partition_dichotomy`, which covers the explicit integer encoding, its entry bounds, the logarithmic Pell search, and the promise form including D-semistability. It depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The polynomial certificates enter the formal proof as ring identities and are checked by the kernel. Section 7 gives the correspondence. What remains informal is the complexity-theoretic packaging in Corollary 5.2: the running-time analysis and the handling of trivial instances.

Organization.

- Section 2: definitions, quartic Routh–Hurwitz facts, absorption and the safety theorem.
- Section 3: the core family, its window and the two certificates.
- Section 4: the proof of the reduction.
- Section 5: the integer encoding and complexity consequences.
- Section 6: a worked example.
- Section 7: the formal verification.
- Section 8: related work and open problems.

2 Preliminaries

2.1 Scalings and decision problems

For $A \in \mathbb{R}^{N\times N}$ and a rate vector $d \in (0,\infty)^N$ put $A^{(d)} = A\,\mathrm{diag}(d)$. It has the same eigenvalues as $\mathrm{diag}(d)A$. The matrix A is:

- *D-stable* if every $A^{(d)}$ is Hurwitz;
- *strictly D-unstable* if some $A^{(d)}$ has an eigenvalue with positive real part;
- *D-semistable* if it is not strictly D-unstable, that is, if no $A^{(d)}$ with $d > 0$ has an eigenvalue with positive real part.

D-stable matrices are D-semistable. All three notions are invariant under multiplication of A by a positive scalar c, since $(cA)^{(d)} = A^{(cd)}$. We use the Hurwitz convention throughout; parts of the literature, e.g. [29], state results for positive stability of $-A$.

With *inverse rates* $\theta = 1/d$ (componentwise), a number z is an eigenvalue of $\mathrm{diag}(d)A$ if and only if the pencil $z\,\mathrm{diag}(\theta) - A$ is singular. We often write eigenvector equations in this pencil form.

The decision problems are:

- D-Stability: given $A \in \mathbb{Q}^{N\times N}$ in binary, is A D-stable?
- D-Semistability: given the same input, is A D-semistable?
- D-Instability: given the same input, is A strictly D-unstable?
- Partition: given positive integers $w_1, \dots, w_m$, is there $S \subseteq [m]$ with $w(S) = w([m]\setminus S)$? Here $w(S) := \sum_{j\in S} w_j$.

2.2 Monic quartics

For $a = (a_1, \dots, a_4) \in \mathbb{R}^4$ put $\Delta(a) = a_1a_2a_3 - a_3^2 - a_1^2a_4$. This is the third Hurwitz determinant of $s^4 + a_1s^3 + a_2s^2 + a_3s + a_4$.

Lemma 2.1 (Quartic Routh–Hurwitz)**.** *Let $a_1, a_2, a_3, a_4 > 0$, and let $\chi(s) = s^4 + a_1s^3 + a_2s^2 + a_3s + a_4$.*

(a) If $\Delta(a) > 0$, every root has negative real part.

(b) If $\Delta(a) < 0$, some root has positive real part.

(c) If $i\omega$ with $\omega \in \mathbb{R}\setminus\{0\}$ is a root, then $a_3 = a_1\omega^2$ and $\Delta(a) = 0$.

Proof. (c) The imaginary part of $\chi(i\omega) = 0$ gives $\omega(a_3 - a_1\omega^2) = 0$. The real part gives $a_4 = a_2\omega^2 - \omega^4$. Substituting both into Δ gives 0.

(a) Here $a_3(a_1a_2 - a_3) = \Delta + a_1^2 a_4 > 0$. So all Hurwitz determinants are positive, and the Routh–Hurwitz criterion [18] applies.

(b) The quartic is not Hurwitz, by the same criterion. Its roots cannot be 0, since $a_4 > 0$, and cannot be imaginary, by (c). Hence some root has positive real part. □

2.3 Star matrices, absorption and safety

Let $B \in \mathbb{R}^{n\times n}$ be a core whose last coordinate n is the port. Let $r \in (0,\infty)^m$, $m \geq 1$, be loads with total $G = \sum_j r_j$. Define

$$A[r] = \begin{pmatrix} B & \mathbf{e}_n r^{\mathsf{T}} \\ \mathbf{1}\mathbf{e}_n^{\mathsf{T}} & -I_m \end{pmatrix}, \qquad B[X] = B + X\mathbf{e}_n\mathbf{e}_n^{\mathsf{T}} \quad (X \in \mathbb{R}).$$

Attachment j obeys $\dot{u}_j = d_{n+j}(x_n - u_j)$, and the port equation receives $\sum_j r_j u_j$. For $z \in \mathbb{C}$ with $\operatorname{Re} z \geq 0$, $r > 0$ and $t > 0$ put

$$\nu = |1 + zt|^2, \qquad \xi(z,r,t) = \frac{r(1 + 2t\operatorname{Re} z)}{\nu}, \qquad \alpha(z,r,t) = \frac{rt}{\nu}.$$

Lemma 2.2 (Absorption)**.** $\dfrac{r}{1+zt} + z\,\alpha(z,r,t) = \xi(z,r,t)$, *and* $\alpha > 0$. *If* $z \neq 0$ *then* $0 < \xi < r$; *if* $z = 0$ *then* $\xi = r$.

Proof. We have $r/(1 + zt) = r(1 + \bar{z}t)/\nu$. Adding zrt/ν gives $r(1 + 2t\operatorname{Re} z)/\nu$. Also $r - \xi = r|z|^2t^2/\nu$. □

Lemma 2.3 (Elimination)**.** *Let d be a rate vector for $A[r]$, with core inverse rates $\theta_i = 1/d_i$ and attachment inverse rates $t_j = 1/d_{n+j}$. Suppose $A[r]^{(d)}$ has an eigenvalue z with $\operatorname{Re} z \geq 0$. Put $X = \sum_j \xi(z, r_j, t_j)$ and $\Lambda = \sum_j \alpha(z, r_j, t_j)$. Then z is an eigenvalue of $B[X]^{(d')}$, where $d'_i = d_i$ for $i < n$ and $d'_n = 1/(\theta_n + \Lambda)$.*

Proof. Write the eigenvector in pencil form as (x, u). The attachment rows give $u_j = x_n/(1 + zt_j)$, and $1 + zt_j \neq 0$ because $\operatorname{Re} z \geq 0$. In particular $x \neq 0$, since otherwise $u = 0$.

The port row receives $\sum_j r_j u_j = x_n \sum_j r_j/(1 + zt_j) = (X - z\Lambda)x_n$ by Lemma 2.2. Moving $-z\Lambda x_n$ to the left turns the port inverse rate θ_n into $\theta_n + \Lambda$ and the load into X. □

Lemma 2.4 (Reverse absorption)**.** *Let z be an eigenvalue of $B[X]^{(d)}$ with $\operatorname{Re} z > 0$, and put $\theta = 1/d$. Suppose $t \in (0,\infty)^m$ satisfies $\sum_j \xi(z, r_j, t_j) = X$ and $\sum_j \alpha(z, r_j, t_j) < \theta_n$. Then z is an eigenvalue of a positive scaling of $A[r]$, namely the one with inner inverse rates θ_i $(i < n)$, port inverse rate $\theta_n - \sum_j \alpha_j$ and attachment inverse rates t_j. In particular $A[r]$ is strictly D-unstable.*

Proof. Take the static eigenvector x, set $u_j = x_n/(1 + zt_j)$, and reverse the computation of Lemma 2.3. □

Theorem 2.5 (Static-interval safety; proved here, cf. [13])**.** *If $B[X]$ is D-stable for every $X \in (0, G)$ and $\det B[G] \neq 0$, then $A[r]$ is D-stable.*

Proof. Let z be an eigenvalue of some $A[r]^{(d)}$ with $\operatorname{Re} z \geq 0$. If $z \neq 0$, Lemma 2.3 makes z an eigenvalue of a positive scaling of $B[X]$ with $0 < X < G$, which is impossible. If $z = 0$, the attachment rows force $u_j = x_n$, so $B[G]x = 0$ with $x \neq 0$, which contradicts $\det B[G] \neq 0$. □

Lemma 2.6 (Continuation)**.** *Let $\mathcal{P}$ be a connected topological space and $\Phi\colon \mathcal{P} \to \mathbb{R}^{N\times N}$ continuous. Suppose no $\Phi(\pi)$ has an eigenvalue on the imaginary axis, and some $\Phi(\pi_0)$ is Hurwitz. Then every $\Phi(\pi)$ is Hurwitz.*

Proof. The Hurwitz matrices form an open set. So do the matrices with an eigenvalue of positive real part: a monic polynomial that is small at z has a root near z. The two sets are disjoint, and by hypothesis they cover $\Phi(\mathcal{P})$. □

3 The core family

Definition 3.1 (Core)**.** For $\rho, h \in \mathbb{R}$ put $\gamma = (16 + 12\rho)/5$ and

$$B(\rho, h) = \begin{pmatrix} -1 & -\frac{2}{5} & -\frac{2}{5} & -1 \\ \gamma & -1 & -\frac{2}{5} & -1 \\ \gamma & \gamma & -1 & -1 \\ -1 & -1 & -1 & -h \end{pmatrix},$$

with the port at coordinate 4. Loading the port moves along the family: $B(\rho, h)[X] = B(\rho, h - X)$.

3.1 Symmetric minors and the Hurwitz decomposition

The principal minors of $-B(\rho, h)$ are equal within each order and type:

$$\begin{aligned} \text{inner:} \quad & 1, \qquad e_2 = \tfrac{57+24\rho}{25}, \qquad e_3 = \tfrac{1053+1080\rho+288\rho^2}{125} && \text{(orders 1, 2, 3)}, \\ \text{with the port:} \quad & h, \qquad m_2 = h - 1, \qquad m_3 = \tfrac{(57+24\rho)h}{25} - \tfrac{24+12\rho}{5} && \text{(orders 1, 2, 3)}, \\ & m_4 = \det(-B) = e_3 h - \tfrac{513+540\rho+144\rho^2}{25} && \text{(order 4)}. \end{aligned}$$

Let the inner rates be $d_1, d_2, d_3 > 0$ and the port rate be $v > 0$. Let $\sigma_1, \sigma_2, \sigma_3$ be the elementary symmetric functions of d_1, d_2, d_3, and

$$U = \sigma_1\sigma_2 - 9\sigma_3 = d_1(d_2 - d_3)^2 + d_2(d_1 - d_3)^2 + d_3(d_1 - d_2)^2 \geq 0.$$

The characteristic polynomial of $B(\rho, h)^{((d_1,d_2,d_3,v))}$ is $s^4 + c_1 s^3 + c_2 s^2 + c_3 s + c_4$ with

$$c_1 = \sigma_1 + hv, \quad c_2 = e_2\sigma_2 + m_2\sigma_1 v, \quad c_3 = e_3\sigma_3 + m_3\sigma_2 v, \quad c_4 = m_4\sigma_3 v. \tag{1}$$

Lemma 3.2 (Hurwitz decomposition)**.** *With $K = 9m_2m_3 - hm_4$,*

$$\Delta(c) = F_0 + F_1 v + F_2 v^2 + h\left(m_2 m_3 U + K\sigma_3\right) v^3,$$

where

$$\begin{aligned} F_0 &= e_3\sigma_3(e_2\sigma_1\sigma_2 - e_3\sigma_3), \\ F_1 &= e_2e_3h\sigma_2\sigma_3 + e_2m_3\sigma_1\sigma_2^2 + e_3m_2\sigma_1^2\sigma_3 - 2e_3m_3\sigma_2\sigma_3 - m_4\sigma_1^2\sigma_3, \\ F_2 &= e_2hm_3\sigma_2^2 + e_3hm_2\sigma_1\sigma_3 - 2hm_4\sigma_1\sigma_3 + m_2m_3\sigma_1^2\sigma_2 - m_3^2\sigma_2^2. \end{aligned}$$

Moreover, with $k_2 = 9e_2 - e_3$, the polynomial $\Psi := c_1(F_0 + F_1 v + F_2 v^2) - k_2 h\sigma_3 v\, c_3$ satisfies

$$\Psi = \sigma_1 F_0 + \left(\sigma_1 F_1 + he_2e_3\sigma_3 U\right) v + \Psi_2 v^2 + hF_2 v^3, \tag{2}$$

where

$$\begin{aligned} \Psi_2 = {} & e_2e_3h^2\sigma_2\sigma_3 + 2e_2hm_3\sigma_1\sigma_2^2 - 9e_2hm_3\sigma_2\sigma_3 + 2e_3hm_2\sigma_1^2\sigma_3 \\ & - e_3hm_3\sigma_2\sigma_3 - 3hm_4\sigma_1^2\sigma_3 + m_2m_3\sigma_1^3\sigma_2 - m_3^2\sigma_1\sigma_2^2. \end{aligned}$$

Proof. Both identities are polynomial identities in $(\rho, h, d_1, d_2, d_3, v)$. □

3.2 The window

The quantity K is quadratic in h:

$$K = k_2h^2 + k_1h + k_0, \qquad k_2 = \tfrac{72(21-4\rho^2)}{125}, \quad k_1 = \tfrac{72(2\rho^2-3\rho-15)}{25}, \quad k_0 = \tfrac{108(\rho+2)}{5}.$$

Here $k_2 = 9e_2 - e_3$ is the same k_2 as in Lemma 3.2. On the parameter range used below $k_2 > 0$. Put $h_v = -k_1/(2k_2)$ and $W = (k_1^2 - 4k_2k_0)/(4k_2^2)$, so that

$$K = k_2((h - h_v)^2 - W). \tag{3}$$

Lemma 3.3 (Discriminant)**.**

$$k_1^2 - 4k_2k_0 = \tfrac{144}{25}\big(\tfrac{2}{5} + \gamma\big)^2(\rho^2 - 3), \qquad \textit{hence} \qquad W = \tfrac{36}{25}\Big(\frac{2/5 + \gamma}{k_2}\Big)^2(\rho^2 - 3).$$

When $k_2 > 0$, the window $\{h : K < 0\}$ is nonempty if and only if $\rho^2 > 3$. It is then the open interval $(h_v - \sqrt{W},\, h_v + \sqrt{W})$.

Remark 3.4 (Not used in the proofs). When the port is infinitely fast, the inner dynamics is governed by the Schur complement $\widehat{B}_h = B_{11} + h^{-1}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ (B_{11} the inner 3×3 block).

- The principal minors of $-\widehat{B}_h$ of orders $1, 2, 3$ are m_2/h, m_3/h and m_4/h.
- Let g_i be the i-th diagonal entry of $-\widehat{B}_h$ and $\hat{g}_i$ its complementary 2×2 principal minor. Cain's criterion [7] compares $(\sum_i \sqrt{g_i\hat{g}_i})^2$, here $9m_2m_3/h^2$, with $\det(-\widehat{B}_h) = m_4/h$.
- So K/h^2 is exactly the Cain gap of $\widehat{B}_h$, and the window is where the port-fast limit fails Cain's test.

Replace the entries $-\frac{2}{5}$ by $-\kappa$ for a general parameter κ. The discriminant then factors as $(\kappa+\gamma)^2((\gamma+7\kappa-6)^2 - 48(1-\kappa)^2)$. So in the two-parameter family, windows are born on the lines $\gamma + 7\kappa - 6 = \pm 4\sqrt{3}(\kappa - 1)$. These lines contain no rational point with $\kappa \neq 1$. Within this family, exactly rational narrow windows therefore come from Diophantine approximations of $\sqrt{3}$.

3.3 Exact certificates

Let $\mathcal{B} = \{(\rho, h) : \frac{17}{10} \le \rho \le \frac{7}{4},\ 3 \le h \le 5\}$.

Proposition 3.5 (Certificates)**.** *Let $(\rho, h) \in \mathcal{B}$.*

(a) $e_2, e_3, k_2, m_2, m_3, m_4 > 0$, and $F_0 > 0$ for all $d_1, d_2, d_3 > 0$.

(b) $F_1, F_2, \Psi_2 \ge 0$ for all $d_1, d_2, d_3 \ge 0$.

(c) $\frac{7}{2} \le h_v \le \frac{9}{2}$ and $\frac{2}{5} + \gamma \le \frac{8}{5}k_2$. Consequently $W \le \frac{2304}{625}(\rho^2 - 3)$ whenever $\rho^2 \ge 3$.

Proof. Parts (a) and (c) are elementary inequalities for $\rho \in [\frac{17}{10}, \frac{7}{4}]$ and $h \ge 3$. For F_0 use $e_2\sigma_1\sigma_2 - e_3\sigma_3 = e_2U + k_2\sigma_3$. For the consequence in (c), use Lemma 3.3 and $\frac{36}{25}(\frac{8}{5})^2 = \frac{2304}{625}$.

For (b), each of F_1, F_2, Ψ_2 is symmetric in d_1, d_2, d_3, so we may assume $d_1 \ge d_2 \ge d_3$. We substitute

$$d_1 = y_1 + y_2 + y_3, \quad d_2 = y_2 + y_3, \quad d_3 = y_3 \qquad (y_i \ge 0),$$

parametrize the box by $\rho = (34\beta_0 + 35\beta_1)/20$ and $h = 3\eta_0 + 5\eta_1$ with $\beta_0 + \beta_1 = \eta_0 + \eta_1 = 1$, and homogenize in (β_0, β_1) and (η_0, η_1). The result is a polynomial H in $(\beta_0, \beta_1, \eta_0, \eta_1, y_1, y_2, y_3)$ that agrees with the given function on the box.

- For F_1 it has degree 3 in β, degree 1 in η and 144 terms.
- For F_2 the degrees are $(2, 2)$ and there are 126 terms.

- For Ψ_2 the degrees are $(3, 2)$ and there are 240 terms.

Every coefficient is nonnegative. The coefficients are the Bernstein coefficients of the function on the box, times binomial factors. Hence the function is nonnegative there for nonnegative y_i. □

Remark 3.6. The certificate is not vacuous: the same procedure applied on the box $h \in [2, \frac{5}{2}]$ produces negative coefficients. On that box F_1, F_2 and Ψ_2 all take negative values at explicit rational points; the determinant crossing $m_4 = 0$ occurs at $h \approx 2.48$.

3.4 Static classification and contact bound

Proposition 3.7 (Static classification)**.** *Let $(\rho, h) \in \mathcal{B}$. If $K(\rho, h) \geq 0$ then $B(\rho, h)$ is D-stable. If $K(\rho, h) < 0$ then $B(\rho, h)$ is strictly D-unstable.*

Proof. By Proposition 3.5(a), all $c_i > 0$ at positive rates.

If $K \geq 0$, every term of Lemma 3.2 is nonnegative and $F_0 > 0$. So $\Delta(c) > 0$ and Lemma 2.1(a) applies.

If $K < 0$, take $d_1 = d_2 = d_3 = 1$, so that $U = 0$ and $\sigma_3 = 1$. For $v > \max\{1, (F_0 + F_1 + F_2)/(h|K|)\}$ we get $\Delta(c) \leq (F_0 + F_1 + F_2)v^2 + hKv^3 < 0$, and Lemma 2.1(b) applies. □

For a scaling with port rate v and an imaginary eigenvalue $i\omega$, call ω/v the *normalized port inverse rate*. It is the port inverse rate $1/v$ after the frequency is normalized to one.

Proposition 3.8 (Contact bound)**.** *Let $(\rho, h) \in \mathcal{B}$. Suppose a scaling of $B(\rho, h)$ with inner rates $d_1, d_2, d_3 > 0$ and port rate $v > 0$ has an eigenvalue $i\omega$ with $\omega \neq 0$. Then*

$$k_2 \, (\omega/v)^2 \leq -K(\rho, h), \qquad \textit{equivalently} \qquad (\omega/v)^2 \leq W - (h - h_v)^2.$$

Proof. By Lemma 2.1(c), $c_3 = c_1\omega^2$ and $\Delta(c) = 0$. Lemma 3.2 and Proposition 3.5 then give

$$-Kh\sigma_3 v^3 = F_0 + F_1 v + F_2 v^2 + hm_2 m_3 U v^3 \geq F_0 + F_1 v + F_2 v^2.$$

Every term on the right of (2) is nonnegative, so $\Psi \geq 0$, that is,

$$k_2 h\sigma_3 v \, c_3 \leq c_1(F_0 + F_1 v + F_2 v^2) \leq -Kh\sigma_3 v^3 c_1.$$

Substituting $c_3 = c_1\omega^2$ and dividing by $c_1 h\sigma_3 v > 0$ gives the claim. □

The contact bound is the main technical ingredient. A single polynomial inequality, $\Psi \geq 0$, places the height of every contact of the static core under the semicircle that spans the window.

4 The reduction

Definition 4.1 (The matrix $A(w)$)**.** Let $w \in \mathbb{N}_{>0}^m$, $m \geq 1$, with $\sum_j w_j = 2T$. Let (p, q) be positive integers with $p^2 - 3q^2 = 1$ and $q \geq \max(8T, 4)$. Put $\rho = p/q$ and $h_0 = h_v(\rho) + \frac{1}{2}$, and define the loads $r_j = w_j/(2T)$, so that $G = 1$. Then

$$A(w) = \begin{pmatrix} B(\rho, h_0) & \mathbf{e}_4 r^{\mathsf{T}} \\ \mathbf{1}\mathbf{e}_4^{\mathsf{T}} & -I_m \end{pmatrix} \in \mathbb{Q}^{(m+4)\times(m+4)}.$$

Lemma 4.2 (Pell parameters)**.** *Under Definition 4.1:*

(a) $\rho^2 = 3 + q^{-2}$ *and* $\frac{17}{10} \leq \rho \leq \frac{7}{4}$*;*

(b) $W(\rho') \leq \frac{2304}{625} q^{-2} \leq \frac{\lambda^2}{4}$ *for every* $\rho' \in [\frac{17}{10}, \rho]$*, where* $\lambda = 1/(2T)$*;*

(c) $W(\rho) > 0$ *and* $W(\frac{17}{10}) < 0$.

Proof. (a) $\rho^2 = (3q^2+1)/q^2$. Since $q \geq 4$, $\rho^2 \leq \frac{49}{16}$, and $\rho^2 > 3 > (\frac{17}{10})^2$.

(b) If $\rho'^2 \geq 3$, Proposition 3.5(c) gives $W(\rho') \leq \frac{2304}{625}(\rho'^2 - 3) \leq \frac{2304}{625}(\rho^2 - 3)$. If $\rho'^2 < 3$ then $W(\rho') < 0$ by Lemma 3.3. Finally $4 \cdot \frac{2304}{625} q^{-2} \leq \frac{1}{4T^2}$ holds when $q^2 \geq \frac{36864}{625} T^2$, which follows from $q \geq 8T$.

(c) Use Lemma 3.3 and $(\frac{17}{10})^2 < 3 < \rho^2$. □

Theorem 4.3. *Let* $A = A(w)$ *be as in Definition 4.1.*

(i) If $w(S) = T$ *for some* S*, then* A *is strictly D-unstable.*

(ii) If $w(S) \neq T$ *for every* S*, then* A *is D-stable.*

Along the load ray the static matrices are $B(\rho, h_0 - X)$ with $X \in [0,1]$. Their load points $h_0 - X = h_v + \frac{1}{2} - X$ lie in $[h_v - \frac{1}{2}, h_v + \frac{1}{2}] \subseteq [3,5]$ by Proposition 3.5(c), so all of them lie in $\mathcal{B}$. The subset sum $r(S) = w(S)/(2T)$ corresponds to the load point $h_v + \lambda(T - w(S))$.

4.1 Instability: exact realization of a subset

Lemma 4.4 (Subset realization)**.** *Let* $\operatorname{Re} z > 0$*, let* $0 < X < G$ *and* $\theta_n > 0$*, and let loads* $r_j > 0$ *with* $\sum_j r_j = G$ *and a subset* S *with* $r(S) = X$ *be given; thus* $\emptyset \neq S \neq [m]$*. Then there are* $t \in (0,\infty)^m$ *with* $\sum_j \xi(z, r_j, t_j) = X$ *and* $\sum_j \alpha(z, r_j, t_j) < \theta_n$.

Proof. Choose $\delta > 0$ with $G\delta + G|z|^2\delta^2/(2\operatorname{Re} z) < \theta_n$ and $G|z|^2\delta^2 < G - X$. Put $f = \xi(z, 1, \delta) \in (0,1)$. From the formula for $r - \xi$ in Lemma 2.2, $1 - f \leq |z|^2\delta^2$. Then $\phi := X(1-f)/(G-X)$ lies in $(0,1)$, since $X(1-f) \leq X|z|^2\delta^2 < G - X$.

The map $t \mapsto \xi(z, 1, t)$ is continuous on $(0,\infty)$, with limits 1 at 0 and 0 at ∞. So some $\tau > 0$ has $\xi(z, 1, \tau) = \phi$; it is also given by an explicit quadratic formula. Set $t_j = \delta$ for $j \in S$ and $t_j = \tau$ otherwise. By the homogeneity $\xi(z, r, t) = r\,\xi(z, 1, t)$,

$$\textstyle\sum_j \xi_j = Xf + (G - X)\phi = X.$$

For the lags, $\alpha(z, r, t) \leq rt$ and $\alpha(z, r, t) \leq \xi(z, r, t)/(2\operatorname{Re} z)$. Hence

$$\textstyle\sum_j \alpha_j \leq X\delta + (G - X)\phi/(2\operatorname{Re} z) \leq G\delta + G|z|^2\delta^2/(2\operatorname{Re} z) < \theta_n.$$ □

Proof of Theorem 4.3 (i). Here $X = r(S) = \frac{1}{2} \in (0,1)$, and the load point is $h_0 - \frac{1}{2} = h_v$. By Lemma 4.2(c) and (3), $K(\rho, h_v) = -k_2 W < 0$. By Proposition 3.7, some scaling of $B(\rho, h_v) = B(\rho, h_0)[\frac{1}{2}]$ has an eigenvalue z with $\operatorname{Re} z > 0$. Lemmas 4.4 and 2.4 make z an eigenvalue of a positive scaling of A. □

4.2 Stability: no contacts, then continuation

Lemma 4.5 (Rounding bound)**.** *Let* $z = i\omega$*,* $r \in (0,\infty)^m$ *and* $t \in (0,\infty)^m$*, and put* $\xi_j = \xi(z, r_j, t_j)$ *and* $\alpha_j = \alpha(z, r_j, t_j)$*. Then* $\omega^2\alpha_j^2 = \xi_j(r_j - \xi_j)$ *for each* j*. Moreover there is a set* S *with*

$$\textstyle\left|\sum_j \xi_j - r(S)\right| \leq |\omega| \sum_j \alpha_j.$$

Proof. Here $\xi_j = r_j/(1 + \omega^2 t_j^2)$ and $\alpha_j = r_j t_j/(1 + \omega^2 t_j^2)$, which gives the identity. Let $S = \{j : 2\xi_j \geq r_j\}$. For every j, $|\xi_j - r_j \mathbf{1}_S(j)| = \min(\xi_j, r_j - \xi_j) \leq \sqrt{\xi_j(r_j - \xi_j)} = |\omega|\alpha_j$. Sum over j and use the triangle inequality. □

Proposition 4.6 (No imaginary-axis eigenvalues)**.** *Let $\rho' \in [\frac{17}{10}, \frac{7}{4}]$ and $h'_0 = h_v(\rho') + \frac{1}{2}$. Let loads $r_j > 0$ $(m \geq 1)$ have total 1, and let $\lambda > 0$ satisfy $4W(\rho') \leq \lambda^2$ and $|r(S) - \frac{1}{2}| \geq \lambda$ for every S. Then no positive scaling of $A' = \begin{pmatrix} B(\rho', h'_0) & \mathbf{e}_4 r^{\mathsf{T}} \\ \mathbf{1}\mathbf{e}_4^{\mathsf{T}} & -I \end{pmatrix}$ has an eigenvalue on the imaginary axis.*

Proof. Suppose z with $\operatorname{Re} z = 0$ is an eigenvalue of $A'^{(d)}$. By Lemma 2.3, z is an eigenvalue of a positive scaling of $B(\rho', h)$ with $h = h'_0 - X$. Here $X = \sum_j \xi_j \in (0, 1]$, so $(\rho', h) \in \mathcal{B}$. The port inverse rate is $\theta_4 + \Lambda$, with $\Lambda = \sum_j \alpha_j \geq 0$.

If $z = 0$, then $c_4 = 0$, contradicting Proposition 3.5(a). If $K(\rho', h) \geq 0$, Proposition 3.7 contradicts $\operatorname{Re} z = 0$.

So assume $z = i\omega$ with $\omega \neq 0$ and $K(\rho', h) < 0$. Then $(X - \frac{1}{2})^2 = (h - h_v)^2 < W$, and Proposition 3.8 gives

$$\omega^2(\theta_4 + \Lambda)^2 \leq W - (X - \tfrac{1}{2})^2 \leq W \leq \lambda^2/4. \tag{4}$$

In particular $|X - \frac{1}{2}| < \sqrt{W} \leq \lambda/2$. Take S from Lemma 4.5:

$$|\omega|\Lambda \;\geq\; |X - r(S)| \;\geq\; |r(S) - \tfrac{1}{2}| - |X - \tfrac{1}{2}| \;>\; \lambda - \lambda/2 = \lambda/2.$$

Hence $\omega^2\Lambda^2 > \lambda^2/4$. Since $\theta_4 > 0$, $\omega^2(\theta_4 + \Lambda)^2 > \omega^2\Lambda^2 > \lambda^2/4$, which contradicts (4). □

Proof of Theorem 4.3 (ii). The w_j are integers and $w(S) \neq T$, so every S satisfies $|r(S) - \frac{1}{2}| = |w(S) - T|/(2T) \geq 1/(2T) = \lambda$. Fix a rate vector d and consider the path

$$\Phi(\rho') = A(\rho')^{(d)}, \qquad A(\rho') = \begin{pmatrix} B(\rho', h_v(\rho') + \frac{1}{2}) & \mathbf{e}_4 r^{\mathsf{T}} \\ \mathbf{1}\mathbf{e}_4^{\mathsf{T}} & -I \end{pmatrix}, \qquad \rho' \in [\tfrac{17}{10}, \rho].$$

It is continuous because $k_2 > 0$ on this interval. By Lemma 4.2(b), Proposition 4.6 applies at every ρ', so no $\Phi(\rho')$ has an imaginary-axis eigenvalue.

At $\rho' = \frac{17}{10}$ we have $W < 0$ by Lemma 4.2(c). So $K > 0$ at every load point, and every static matrix $B[X]$, $X \in [0, 1]$, is D-stable by Proposition 3.7. In particular $\det B[1] \neq 0$. Theorem 2.5 makes $A(\frac{17}{10})$ D-stable, so $\Phi(\frac{17}{10})$ is Hurwitz.

Lemma 2.6 makes $\Phi(\rho) = A^{(d)}$ Hurwitz. Since d was arbitrary, A is D-stable. □

Remark 4.7. The stable direction uses the elimination lemma and the elementary rounding bound, not the fixed-partition theory of [13]. The safety theorem is used only at the window-free base point, and the only topological input is the continuation in ρ.

5 Integer encoding and complexity

Proposition 5.1 (Encoding)**.** *Let (p, q) be a Pell pair ($p^2 - 3q^2 = 1$, $q \geq 1$) and $\sum_j w_j = 2T$.*

(a) $h_0 = h_v(p/q) + \frac{1}{2} = \dfrac{54q^2 + 15pq - 14}{2(9q^2 - 4)}$.

(b) Put $L = 10q(9q^2 - 4)T$ and $E = 9q^2 - 4$. Then $M(w) = L \cdot A(w)$ is an integer matrix with core block

$$\begin{pmatrix} -L & -4qET & -4qET & -L \\ 8TE(3p + 4q) & -L & -4qET & -L \\ 8TE(3p + 4q) & 8TE(3p + 4q) & -L & -L \\ -L & -L & -L & -5Tq(54q^2 + 15pq - 14) \end{pmatrix}.$$

Its other nonzero entries are $M_{4,4+j} = 5qEw_j$, $M_{4+j,4} = L$ and $M_{4+j,4+j} = -L$.

(c) *Since $p \le 2q$ and $w_j \le 2T$, every entry satisfies $|M_{ij}| \le 720\, q^3 T$.*

(d) *The recursion $(p_0, q_0) = (2, 1)$, $(p_{k+1}, q_{k+1}) = (2p_k + 3q_k,\ p_k + 2q_k)$ produces Pell pairs with $q_k \le p_k \le 2q_k$ and $q_k \ge 3^k$. The first index with $q_k \ge N_0$ is at most $\lfloor \log_3 N_0 \rfloor + 1$, and at that index $q_k \le 4N_0 + 1$.*

Proof. (a) Write h_v as a rational function of $\rho = p/q$. The difference of the two sides has numerator $30q(2p + 3q)(p^2 - 3q^2 - 1)$.

(b) Direct computation.

(c) Use $p \le 2q$, $E \le 9q^2$ and $w_j \le 2T$. The largest entry is $8TE(3p + 4q) \le 720q^3T$.

(d) Each step preserves $p^2 - 3q^2 = 1$. From $q^2 < p^2 = 3q^2 + 1 \le 4q^2$ we get $q \le p \le 2q$. Hence $3q_k \le q_{k+1} \le 4q_k$. The first bound gives $q_k \ge 3^k$, so the index is at most $\lfloor \log_3 N_0 \rfloor + 1$. The second bound gives $q_k \le 4q_{k-1} < 4N_0$ at the first index $k \ge 1$, and $q_0 = 1 \le 4N_0 + 1$. □

Corollary 5.2 (Complexity)**.** *The following map is computable in time polynomial in $m + \sum_j \log_2(w_j + 1)$:*

- *if $\sum_j w_j$ is odd, a no-instance, output the D-stable matrix (-1);*
- *if $m = 0$, a yes-instance, output the strictly D-unstable matrix (1);*
- *otherwise put $T = \frac{1}{2}\sum_j w_j$ and $N_0 = \max(8T, 4)$, take the first Pell pair of Proposition 5.1(d) with $q \ge N_0$, and output $M(w)$.*

Yes-instances go to strictly D-unstable matrices and no-instances to D-stable ones. Consequently:

- Partition *reduces in polynomial time to* D-Instability*, which is therefore NP-hard;*
- *the complement of* Partition *reduces to* D-Stability *and to* D-Semistability*, which are therefore coNP-hard.*

Proof. The Pell search takes $O(\log T)$ steps on integers of $O(\log T)$ bits. The matrix has $(m + 4)^2$ entries, each an integer of magnitude at most $720(4N_0 + 1)^3 T = O(T^4)$. So the output has $O(m^2 \log T)$ bits and is computed with $O(m^2)$ arithmetic operations on $O(\log T)$-bit integers. Positive scaling preserves all three properties, so Theorem 4.3 applies to $M(w)$. □

6 A worked example

Take $T = 2$, so $N_0 = 16$ and the Pell search stops at $(p, q) = (97, 56)$. Then

$$\rho = \tfrac{97}{56}, \quad \gamma = \tfrac{103}{14}, \quad h_v = \tfrac{22259}{5644}, \quad h_0 = \tfrac{25081}{5644}, \quad \lambda = \tfrac{1}{4}.$$

The window is the open interval

$$\left(\tfrac{133}{34},\ \tfrac{330}{83}\right) = \left(h_v - \tfrac{181}{5644},\ h_v + \tfrac{181}{5644}\right),$$

of half-width $181/5644 \approx 0.032 < \lambda/2$. The load points of the subset sums $s \in \{0, \dots, 4\}$ are $h_0 - s/4$:

$$\tfrac{25081}{5644}, \quad \tfrac{11835}{2822}, \quad \tfrac{22259}{5644}, \quad \tfrac{5212}{1411}, \quad \tfrac{19437}{5644}.$$

Only $s = 2$ gives a point inside the window, and it lies at the centre (Figure 2). The integer version $M(w) = L\, A(w)$ has $L = 31\,606\,400$.

No-instance. For $w = (1, 3)$ the subset sums are $\{0, 1, 3, 4\}$, so the 6×6 matrix $A(1, 3)$ is D-stable by Theorem 4.3.

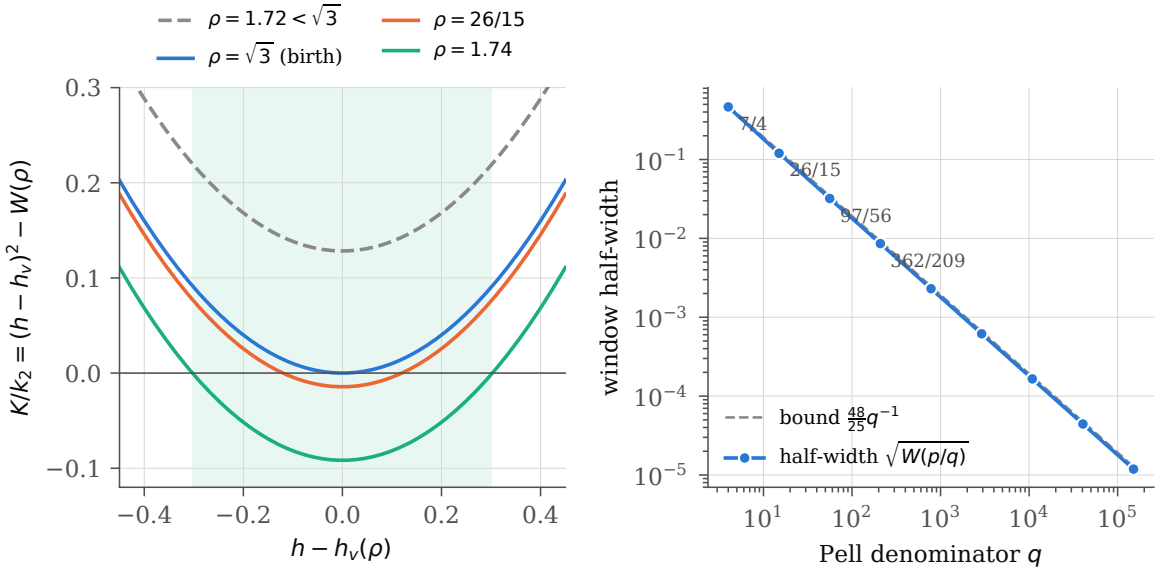


Figure 1: Left: $K/k_2 = (h - h_v)^2 - W$ against $h - h_v$ for $\rho \in \{1.72, \sqrt{3}, 26/15, 1.74\}$. The four curves are vertical translates of one parabola. For $\rho < \sqrt{3}$ there is no window; at $\rho = \sqrt{3}$ it is born as a double root; for $\rho > \sqrt{3}$ it is the interval where the curve is negative, shaded for $\rho = 1.74$. Right: the half-width $\sqrt{W}$ of the window at the Pell parameters $\rho = p/q$, against q. It decays like q^{-1}. The dashed line is the bound $\frac{48}{25}q^{-1}$ implied by Proposition 3.5(c); it exceeds the actual half-width by less than 7%.

Yes-instance. For $w = (1, 1, 2)$ the subset $S = \{3\}$, which consists of the third weight $w_3 = 2$, sums to T. Theorem 4.3 yields a destabilizing scaling. A simpler one, found by hand following the mechanism of the proof, assigns:

- rate 1 to the three inner coordinates;
- rate $2 \cdot 10^5$ to the port and to the attachment of weight 2;
- rate 10^{-4} to the two attachments of weight 1.

The characteristic polynomial of the scaled 7×7 rational matrix has exactly two roots in the open right half-plane, certified by the exact Routh array. Numerically they are $\approx 2.0 \cdot 10^{-6} \pm 2.24\,i$. The fast attachment and the fast port realize the static load $\frac{1}{2}$ at the window centre, while the slow attachments are nearly invisible. The construction of Lemma 4.4, executed literally, also produces a certified destabilizing scaling, but with far more extreme rate ratios.

7 Formal verification

The development is in `proofs/DStabilityHardness/`. It is included, with every module it imports, in the ancillary files of this article, under `anc/lean/proofs/`. The paper-only root is `Paper.lean`. Its import closure has 52 modules including itself: the 13 modules of Table 1 and 39 library modules that they import directly or indirectly. It does not contain the modules of the follow-up results mentioned below.

- It builds on verified libraries for the companion paper [13], namely absorption, reverse absorption, the safety theorem and spectral continuation, and on a verified classification of positive-coefficient quartics.
- It was compiled with Lean v4.30.0 and Mathlib at commit `c5ea00351c28`, with mani-

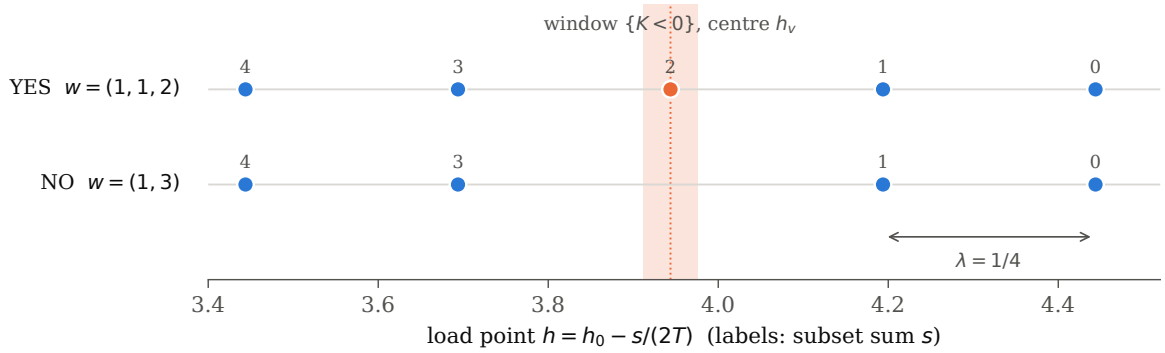


Figure 2: The worked example with $(p, q) = (97, 56)$, $T = 2$, $\lambda = \frac{1}{4}$, on the axis of the load point h. The shaded interval is the window $\{K < 0\}$, of half-width $181/5644$. Its centre h_v is the load point of the subset sum T. Top: $w = (1, 1, 2)$; the sum 2 lands at the centre, so the matrix is strictly D-unstable. Bottom: $w = (1, 3)$; all sums stay at distance $\geq \lambda$ from the centre, so the matrix is D-stable.

fest hash `a8df0b606ec2...`, treating warnings as errors. (This is the SHA-256 of the `lake-manifest.json` of the verification environment; the shipped manifest differs only in the name of the root package.)

- The exported declarations depend only on `propext`, `Classical.choice` and `Quot.sound`.
- No `sorry`, `admit`, `native_decide` or additional axiom occurs.

Positive diagonal scalings act on the right, $A \mapsto A\,\mathrm{diag}(d)$. D-stability, strict D-instability and D-semistability have literal eigenpair semantics over $\mathbb{C}$: the definitions `DStable`, `DUnstable` and `DNonUnstable`.

How the certificates are checked. The three polynomials of Proposition 3.5(b) are handled identically.

1. A script generates their homogenized forms as explicit Lean definitions, in blocks of at most 24 terms.
2. Lean proves, by `ring`, that each function under the substitution equals its form.
3. Writing every atom as a square, `positivity` proves each block nonnegative.

An eight-case sorting argument extends the ordered case to all nonnegative rates.

The formal main theorem. Let the weights w be positive integers on a nonempty finite index type, with $\sum w = 2T$. The theorem `partition_dichotomy` states the following. Let $N_0 = \max(8T, 4)$, let k be the index returned by `pellIndex` for N_0, and let (p, q) be the k-th term of `pellSeq`. Then:

- $k \leq \lfloor \log_3 N_0 \rfloor + 1$ and $q \leq 4N_0 + 1$;
- every entry of the integer matrix `hardIntMatrix` $p\,q\,w\,T$ is bounded by $720q^3T$;
- if no subset of w sums to T, its real cast is D-stable and D-semistable;
- if some subset sums to T, its real cast is strictly D-unstable, and neither D-stable nor D-semistable.

Table 1: Correspondence between the paper and the Lean development. File names are relative to `anc/lean/proofs/`; "companion" abbreviates `DStabilityCharacterization/`, "hardness" abbreviates `DStabilityHardness/`, and "quartic library" is `CollectiveInstability/QuarticClassification`.

Result	File	Declaration(s)
Lemma 2.1	quartic library; hardness `Static`	`matrix_hurwitzStable_of_delta_pos`, `matrix_unstable_of_delta_neg`; `quartic_axis_relations`
Lemma 2.2	companion `Granularity`	`absorption`, `load_bounds`, `shift_pos`
Lemmas 2.3, 2.4	hardness `Window`; companion `FiniteRealization`	`attached_pencil`, `attached_static`, `eigenpair_of_pencil`; `reverse_absorption`
Theorem 2.5, Lemma 2.6	companion `Granularity`, `SpectralContinuation`	`attached_dStable_of_det`; `hurwitzStable_on_preconnected`
(1), $U \geq 0$, Lemmas 3.2, 3.3	hardness `Family`, `Static`, `Window`	`scaled_charpoly`, `U_nonneg`, `delta3_decomposition`, `psi_decomposition`, `Kf_quadratic`, `k2_eq`, `disc_factor`, `Kf_vertex`, `Kf_window`
Proposition 3.5	hardness `Certificates`, `Symmetric`, `Static`, `Window`	`F1_nonneg`, `F2_nonneg`, `Psi2_nonneg`; `e2_pos`, ..., `m4_pos`, `F0_pos`; `hv_bounds`, `Wd_le`
Propositions 3.7, 3.8	hardness `Static`	`static_dStable`, `static_dUnstable`, `contact_bound`
Lemma 4.2	hardness `Reduction`, `Window`, `Paper`	`pell_facts`, `Wd_le`, `Wd_base_neg`
Lemmas 4.4, 4.5; Proposition 4.6	hardness `Reduction`, `Window`	`subset_realization`; `rounding_bound`; `no_axis`
Theorem 4.3	hardness `Reduction`, `Main`	`no_direction`, `hardness_reduction`, `hardness_iff`, `dStability_partition_reduction`
Proposition 5.1	hardness `Encoding`	`h0_formula`, `hardIntMatrix_cast`, `hardIntMatrix_bound`, `pellSeq_pell`, `pellSeq_three_pow`, `pellSeq_p_le`, `pellIndex_le_log`, `pellIndex_q_le`
Scalar invariance; main theorem (promise form)	hardness `Encoding`, `Paper`	`dStable_smul_iff`, `dUnstable_smul_iff`; `dStability_coNP_reduction`, `partition_dichotomy`

What is not formal.

- Remarks 3.4 and 3.6. They are not used in the proofs; their computations are checked in exact arithmetic by `check_paper.py`.
- The complexity-theoretic packaging of Corollary 5.2: that the map is computable in polynomial time, the handling of odd totals and of $m = 0$, and the notion of a Karp reduction. It follows from the explicit formulas, the entry bound and the logarithmic Pell search, all of which are formal.

In complexity theory the running time of an explicit arithmetic map is usually checked by inspection, as here. Mechanized complexity theory exists: Turing-machine programming and the Cook–Levin theorem have been formalized in Coq [16, 17] and in Isabelle [2]. Mathlib provides an interface for polynomial-time Turing computability (`Turing.TM2ComputableInPolyTime`), but even its closure under composition is not yet proved there. A Lean proof of the running time of our reduction would need, on top of that interface, a verified cost model for arithmetic

on binary integers. This is outside the scope of the present paper.

The receipt used for this paper verifies `Paper.lean`. One further declaration is quoted in Section 8, open problem 3, and is not used in any proof: `destabilizing_port_fast`, in the module `TimeScale`. It is verified with a second root, `All.lean`. Its import closure contains four modules beyond those of `Paper.lean`, and it was checked with the same toolchain, Mathlib revision and settings; the declaration depends on the same three axioms.

Exact computation. The script `check_paper.py` in the ancillary files performs 42 checks. In exact rational arithmetic it replays:

- the identities and certificates of Section 3, including the computations of Remark 3.4 and the negative control of Remark 3.6;
- the Pell pairs and bounds, and the integer encoding;
- every number of Section 6, with the exact Routh certificate of the displayed destabilizing scaling;
- the construction of Lemma 4.4, executed literally on small yes-instances and certified by exact Routh arrays.

It also checks that every shipped Lean source has the hash recorded in the verification receipts, and it regenerates the certificate files `Certificates.lean` and `CertificatesTimeScale.lean` byte for byte from the formulas of Lemma 3.2. The script does not rerun Lean. Two checks are floating-point sanity checks and are labelled as such: the 7% statement of Figure 1 and the numerical value of the unstable pair in Section 6.

8 Related work and open problems

Relation to known hardness results.

- *Structured singular values.* Chen, Fan and Yu [10] related D-stability to a condition on the real structured singular value μ of a matrix built from A. Lee and Edgar [31] gave the condition in the non-strict form that characterizes D-stability; see [29] for this history. The recognition problem for real μ is NP-hard [6], but only for general matrices. The hardness does not transfer to the special matrices arising from D-stability: a reduction *to* μ gives an upper bound, not a lower one.
- *Interval matrices.* Poljak and Rohn [34] and Nemirovskii [33] proved that deciding whether every member of an interval matrix is nonsingular, respectively Hurwitz, is intractable; these universal questions are coNP-hard in the terminology used here. In an interval family the entries vary independently over a closed box. In the D-scaling family $\{\mathrm{diag}(d)A\}$ one parameter scales a whole row, and the parameters range over an open cone; after normalization this is the relative interior of a polytope whose vertices are singular (see the next item). We are not aware of a reduction between the two kinds of family. Structurally closest are two bordered constructions, like our star matrices. Nemirovskii's result holds already when the uncertain entries lie in the last row and column only [33]; see also the survey [23]. Blondel and Tsitsiklis [4] reduced PARTITION to the existence of a Hurwitz matrix in an interval family whose uncertain entries also lie in the last row and column. Our reduction is in the same spirit, but it must control arbitrarily large ratios between the rates d_i instead of entries in a closed box.
- *Matrix polytopes.* Gurvits and Olshevsky [21] proved that deciding whether every matrix in the convex hull of given matrices is Hurwitz is intractable (again a coNP-hard universal question in our terminology). Since D-stability is invariant under positive

scalar multiples of d, A is D-stable if and only if every matrix in the relative interior of the polytope $\operatorname{conv}\{\mathbf{e}_i\mathbf{e}_i^{\mathsf{T}}A : 1 \le i \le N\}$ is Hurwitz. For $N \ge 2$ the generating matrices $\mathbf{e}_i\mathbf{e}_i^{\mathsf{T}}A$ have rank at most one and are never Hurwitz. So D-stability concerns the relative interior of a very special polytope, and their result does not imply ours.

- *P-matrices.* Coxson [11] proved that recognizing P-matrices is coNP-complete. In the Hurwitz convention, D-stability implies that $-A$ is a P_0^+-matrix [12], but the converse fails, so no reduction follows.

The survey of complexity results in systems and control by Blondel and Tsitsiklis [5] does not treat D-stability. Our reduction exploits a dynamical mechanism: the subset-sum obstruction of the companion paper [13], made exact by the certificates of Section 3.

Upper bounds and where the hardness lives. Strict D-instability asks for $d > 0$ and z with $\operatorname{Re} z > 0$ and $\det(z - \operatorname{diag}(d)A) = 0$. This is a sentence of the existential theory of the reals. Likewise, D-stability and D-semistability are negations of existential sentences. All three are therefore decidable in polynomial space [8, 35].

For fixed order N the number of variables is fixed. The problems are then decidable in time polynomial in the bit length [35, 36, 37, 3], so, unless P = NP, hardness requires growing dimension.

Our instances need entries of magnitude $\Theta(T^4)$ and exponentially fine windows. Accordingly, the hardness is weak: with unary weights, T is polynomial and PARTITION is easy.

Open problems.

1. *Strong hardness.* Is D-stability coNP-hard for matrices whose entries have polynomially bounded numerators and denominators? A natural route combines several ports, each carrying its own window.
2. *Membership and exact complexity.* Is D-semistability in coNP? Equivalently, is strict D-instability in NP? This would follow, for instance, if every strictly D-unstable rational matrix had a destabilizing scaling of polynomial bit length. At the other end, are these problems $\exists\mathbb{R}$-hard?
3. *Bounded heterogeneity.* The destabilizing scalings in the proof of Theorem 4.3 (i) use rate ratios that grow with T. In the proof the port is faster than the inner coordinates by a factor of order $|K|^{-1} = \Theta(q^2)$ (proof of Proposition 3.7), and the attachment rates of Lemma 4.4 are more extreme still. This is unavoidable for our matrices: every positive scaling of $A(w)$ with an eigenvalue in the closed right half-plane has port rate larger than q^2 times the sum of the three inner rates. (This is the Lean declaration `destabilizing_port_fast`, see Section 7; it is not used in our proofs.) The port rate is at most $\max_i d_i$ and the sum of the inner rates is at least $3\min_i d_i$, so such a scaling has $\max_i d_i/\min_i d_i > 3q^2$. Hence, once $3q^2 \ge R$, every positive scaling of $A(w)$ with $\max_i d_i/\min_i d_i \le R$ is Hurwitz, whatever the answer to the PARTITION instance, and the reduction says nothing about bounded heterogeneity. Is D-stability restricted to scalings with $\max_i d_i/\min_i d_i \le R$ still hard for fixed R?

Data and code availability

The ancillary files of this article (directory `anc/`) contain:

- the Lean sources of both roots and of every module they import, with the toolchain pin, the Mathlib manifest and a `lakefile.lean`;

- summaries of the two verification receipts, listing the SHA-256 hash of every source file and the axioms of the exported declarations;
- the exact replay script `check_paper.py` with its helper module, and the generator of the certificate files;
- a `README.md` with the verification commands.

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